

Total Questions: 35

LEVEL – II

SET - B

Time: 1.5 hr.
Total Marks: 70

SECTION - 1

- This section contains **FIFTEEN (15)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.

SECTION - 2

- This section contains **EIGHT (08)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each correct answer you will get **TWO** marks, if all the options are correct. If the correct answers chosen are less than the actual correct answers, partial marking (basis on weightage) will be given.
- There are **NO NEGATIVE** marks for wrong answers.

SECTION - 3

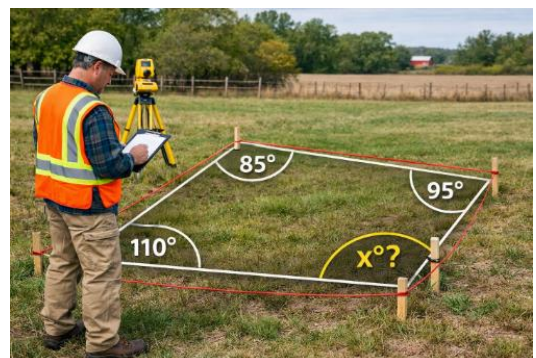
- This section contains **NINE (09)** questions of **PASSAGE TYPE OR COMPREHENSION TYPE**.
- Under each paragraph there will be three questions that has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.

SECTION - 4

- This section contains **THREE (03)** questions. These questions are **MATCHING TYPE BASED QUESTIONS**.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers

Section - 1

1. A land surveyor marks four boundary angles of a field as 85° , 95° , 110° , and x° . He confirms the shape is a quadrilateral and measurements are accurate. The missing angle determines fencing alignment.



- (A) 60° (B) 70°
(C) 75° (D) 80°
2. The total cost of 3 pens and 2 notebooks is ₹34. Another purchase of 5 pens and 2 notebooks costs ₹50. What is the cost of one pen?
- (A) ₹4 (B) ₹8
(C) ₹6 (D) ₹7
3. A treasure is hidden exactly midway between points A(4, -2) and B(10, 6). A student claims the point is (6, 2). Is the claim correct?
- (A) Yes
(B) No, correct point is (7, 2)
(C) No, correct point is (6, 1)
(D) No, correct point is (8, 4)
4. An architect is designing a triangular glass panel for a modern building. The three edges of the panel measure 9 m, 10 m, and 17 m. To estimate the cost of material, she needs to find the exact area of the panel. Since the height is not directly available, she uses Heron's formula.
What is the area of the glass panel?
- (A) 36 m^2 (B) 40 m^2
(C) 44 m^2 (D) 48 m^2
5. Given that $(x+2)$ and $(x-1)$ are factors of the polynomial $x^3 + 10x^2 + mx + n$ the values of m and n are:
- (A) $m=5, n=-3$ (B) $m=7, n=-18$
(C) $m=17, n=-8$ (D) $m=23, n=-14$
6. A surveyor marks a location using a dot on a map. From that point, several straight roads extend in different directions without curvature.

He claims that these roads represent lines, while the dot represents a point. Based on Euclid's definitions, which statement is correct?

- (A) A point has length but no breadth
(B) A line has only length and no breadth
(C) A point can be extended infinitely
(D) A line has fixed endpoints
7. A teacher writes two fractions representing the same shaded portion of a diagram:
 $\frac{3}{4}$ and $\frac{9}{12}$
Which statement is correct?
- (A) They are not equal because denominators differ
(B) They are equivalent rational numbers
(C) Only one is in simplest form
(D) Both are irrational
8. A triangular park has three corner angles such that one angle is twice the second and three times the third. The park designer wants to check if such a triangle is possible. Find the nature of the triangle based on angles.
- (A) Acute triangle
(B) Right triangle
(C) Obtuse triangle
(D) Impossible triangle
9. Two parallel electric wires are cut by a pole forming angles $(3x + 15)^\circ$ and $(3x - 15)^\circ$ on the same side interior. Find x.
- (A) 25 (B) 30
(C) 35 (D) 40
10. If $x - 3$ is a factor of $p(x) = 2x^3 - 5x^2 + kx - 3$, find k.
- (A) -2 (B) 1
(C) 0 (D) -1

11. Three girls Reshma, Salma and Mandip are playing a game by standing on a circle of radius 5m drawn in a park. Reshma throws a ball to Salma, Salma to Mandip, Mandip to Reshma. If the distance between Reshma and Salma and between Salma and Mandip is 6m each, what is the distance between Reshma and Mandip?

- (A) 4.8 m (B) 7.2 m
(C) 9.6 m (D) 10 m

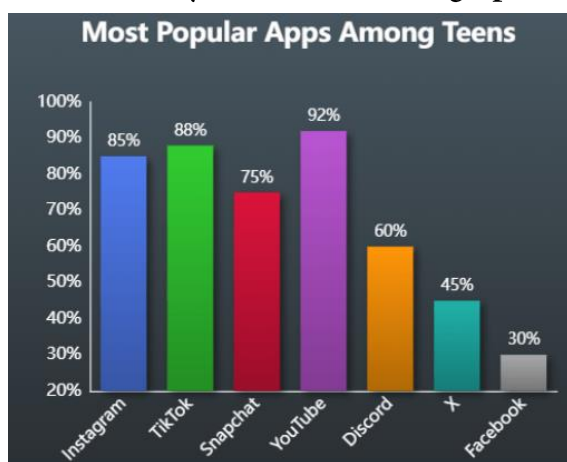
12. Two roads intersect at a junction forming vertically opposite angles. One of the angles is measured as $(5x - 30)^\circ$, while its vertically opposite angle is $(3x + 10)^\circ$. Find x .

- (A) 10 (B) 15
(C) 20 (D) 25

13. A circle is drawn with its center at the origin (0,0). If a point (3, 4) lies on the circumference, which of these points also lies on the circle?

- (A) (0,7) (B) (5,0)
(C) (2,6) (D) (1,4)

14. Which of the following statements is mathematically true based on the graph?



(A) TikTok is twice as popular as X.

(B) The popularity of YouTube and Facebook combined is exactly 120%.

(C) Instagram and Snapchat together have a higher popularity than YouTube and Discord combined.

(D) The Median popularity value of these 7 apps is 88%.

15. A machine outputs a value: $\sqrt{48} + \sqrt{27}$. What is the simplified result?

- (A) $5\sqrt{3}$ (B) $7\sqrt{3}$
(C) $6\sqrt{3}$ (D) $4\sqrt{3}$

Section - 2

16. Two metal rods are equal in length. Both are compared with a third rod and match exactly. Later, both rods are extended by adding equal pieces.

Which conclusions are logically correct?

- (A) Both rods remain equal after extension
(B) Both rods were equal initially
(C) Equality cannot be concluded
(D) Equality is preserved after adding equals

17. A land plot ABCD is fenced such that all four interior angles are measured.

The surveyor confirms it is a closed quadrilateral with no intersecting sides. He wants to verify fundamental properties.

Which of the following are always true?

- (A) Sum of interior angles = 360°
(B) It always has exactly 2 diagonals
(C) Opposite sides are always equal
(D) It can be divided into two triangles

18. A storage container's volume is represented by $V(x) = x^3 + 6x^2 + 11x + 6$. If the dimensions (length, width, height) are linear factors of $V(x)$, which of the following are possible dimensions of the container?

- (A) $(x+1)$ (B) $(x+2)$
(C) $(x+3)$ (D) $(x+6)$

19. A builder uses sets of rods to construct right-angled triangular frames.

He selects combinations that satisfy Pythagoras property.

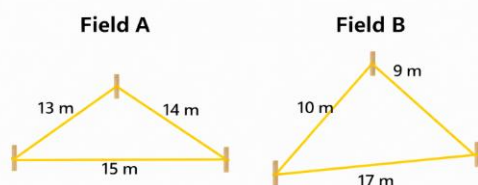
Identify valid triplets.

- (A) (6, 8, 10) (B) (9, 10, 19)
(C) (8, 15, 17) (D) (7, 24, 26)

20. Two triangular fields are given:

Field A $\rightarrow$ sides 13 m, 14 m, 15 m

Field B $\rightarrow$ sides 9 m, 10 m, 17 m



Which are correct?

- (A) Area of Field A is 84 m^2
(B) Area of Field B is 36 m^2
(C) Field A has larger area
(D) Field B is right-angled

21. Two straight roads are constructed parallel to each other. A third road crosses them, forming several angles. The engineer marks one angle as 110° at the intersection.

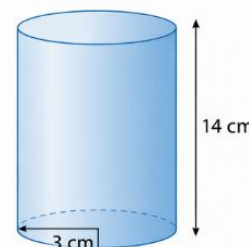
Which of the following statements must be correct?

- (A) The corresponding angle is 110°
(B) The alternate interior angle is 110°

(C) The co-interior angles sum to 180°

(D) The vertically opposite angle is 70°

22. A cylinder has radius 3 cm and height 14 cm. (Use $\pi = 22/7$)



Which are correct?

- (A) Volume = 396 cm^3
(B) Curved surface area = 264 cm^2
(C) Total surface area = 350 cm^2
(D) Diameter = 6 cm

23. The following data shows number of absentees in a class over 10 days:

2, 3, 5, 3, 2, 4, 5, 2, 3, 4

Which statements are correct?

- (A) Mode = 2 and 3 (B) Median = 3
(C) Mean = 3.3 (D) Range = 3

Section - 3

PASSAGE 1:

School Ticket Distribution

A school sells tickets for a function:

student tickets cost ₹30 and adult tickets cost ₹80.

On a particular day, total tickets sold were 50.

The total collection was ₹2600.

Let x be number of student tickets and y be number of adult tickets.

The school needs to verify ticket distribution.



24. Which system of equations represents the situation?

- (A) $x + y = 50$ and $30x + 80y = 2600$
- (B) $x - y = 50$ and $30x + 80y = 2600$
- (C) $x + y = 2600$ and $30x + 80y = 50$
- (D) $30x + 80y = 50$ and $x - y = 2600$

25. If adult tickets were priced ₹10 higher (₹90), keeping same numbers of tickets, the total collection would be:

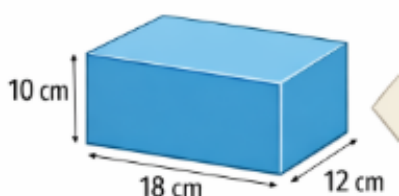
- (A) ₹2800
- (B) ₹2820
- (C) ₹2900
- (D) ₹3100

26. The graphical representation of the equations will be:

- (A) Parallel lines
- (B) Intersecting lines
- (C) Coincident lines
- (D) Perpendicular lines

PASSAGE-2

A carpenter builds a cuboidal box of dimensions $18 \text{ cm} \times 12 \text{ cm} \times 10 \text{ cm}$. He wants to minimize wood usage by comparing it with a cube of same volume. He calculates surface area, diagonal, and edge length for both shapes.



27. Volume of the cuboid is:

- (A) 2160 cm^3
- (B) 2000 cm^3
- (C) 1800 cm^3
- (D) 2400 cm^3

28. Total surface area of the cuboid is:

- (A) 936 cm^2
- (B) 816 cm^2
- (C) 1032 cm^2
- (D) 960 cm^2

29. Total edge length of cuboid is:

- (A) 160 cm
- (B) 120 cm
- (C) 80 cm
- (D) 200 cm

PASSAGE-3

Archaeologists have divided a rectangular excavation site into four quadrants to map artifacts. They use the center of the site as the origin (0, 0).

- **Artifact A** (a gold coin) was found at (-3, 2).
- **Artifact B** (a clay pot) was found at (5, 2).
- **Artifact C** (a stone tool) was found at (-3, -4).

Use this information to answer the following questions.

30. In which quadrant was Artifact A discovered?

- (A) Quadrant I
- (B) Quadrant II
- (C) Quadrant III
- (D) Quadrant IV

31. What is the perpendicular distance of Artifact B from the y-axis?

- (A) 2 units
- (B) 5 units
- (C) 7 units
- (D) 3 units

32. If the archaeologists find Artifact D such that ABCD forms a rectangle, what must be the coordinates of Artifact D?

- (A) (5, -4)
- (B) (-5, 4)
- (C) (3, 4)
- (D) (5, 4)

Section - 4

33. A mathematician simplifies expressions before interpreting results

Column I	Column II
(a) $(2^3 \times 2^2)^2$	(i) 3^7
(b) $(3^2)^3 \times 3^1$	(ii) 2^{10}
(c) $(7^{-1})^2$	(iii) 10^1
(d) $10^6 \div 10^2 \times 10^{-3}$	(iv) $\frac{1}{49}$

- (A) a-i, b-ii, c-iii, d-iv
 (B) a-ii, b-i, c-iv, d-iii
 (C) a-i, b-ii, c-iv, d-iii
 (D) a-iii, b-ii, c-i, d-iv

34. Two locations are connected and midpoint is used for placing a facility

Column I (EndPoints)	Column II (MidPoint)
(a) A(2, 4), B(6, 8)	(i) (4, 2)
(b) C(-2, 6), D(4, 0)	(ii) (4, 6)
(c) E(0, 0), F(8, 4)	(iii) (1, 3)
(d) G(-6, -2), H(2, 6)	(iv) (-2, 2)

- (A) a-i, b-ii, c-iii, d-iv
 (B) a-ii, b-i, c-iii, d-iv
 (C) a-i, b-ii, c-iv, d-iii
 (D) a-ii, b-iii, c-i, d-iv

35. A student is given different solid shapes used in daily life like tanks, ice-cream cones, and balls. Using the given dimensions, calculate the required values and match them with the correct results. (Take $\pi = 22/7$)

Column I (Situations)	Column II (Values)
(a) A cylindrical water tank has radius 7 cm and height 5 cm. Find its volume	(i) 1078 cm^3
(b) A conical sand heap has radius 7 cm and height 21 cm. Find its volume	(ii) 770 cm^3
(c) A spherical ball has radius 7 cm. Find its curved surface area	(iii) 47.1 cm^2
(d) A cone-shaped cap has radius 3 cm and slant height 5 cm. Find its curved surface area	(iv) 616 cm^2

- (A) a-i, b-ii, c-iii, d-iv
 (B) a-ii, b-i, c-iv, d-iii
 (C) a-i, b-ii, c-iv, d-iii
 (D) a-iii, b-ii, c-i, d-iv